EduTerm AI: ASR- and LLM-Based Terminology Extraction and Pedagogical Knowledge Graph (DAG) Evaluation for Multilingual Education

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INTRODUCTION AND MOTIVATION

- **Technical-Jargon Barrier:** Heterogeneous prior knowledge in lecture videos causes severe cognitive overload.
- **Personalization Barrier:** 1-to-1 tutoring improves learning by **2 standard deviations (Bloom 2-sigma)**, but human support is unscalable.
- **Automation Need:** End-to-end ASR–LLM integration extracts terms dynamically and builds scaffolding.

GOAL AND SCOPE (AIM)

Objective: Benchmark Whisper, Qwen3-ASR, Nemotron, local SLMs/LLMs (Qwen 2.5/3.5), and EVREN AI Platform (H200) across accuracy, latency, VRAM, and term F_1 on a 40-video corpus (12 disciplines, 10.1 h) with rigorous double-annotation verification ($N = 115$).

ACOUSTIC PIPELINE AND ARCHITECTURE

- 16 kHz mono audio, 30 s windowing, Whisper `sequential_generate()` (95% storage saving via efficient stream chunking).

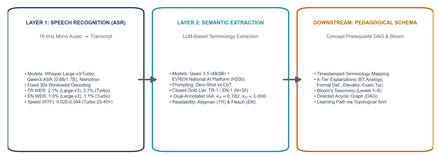


Figure 1: End-to-End Multilayer Architectural Pipeline.

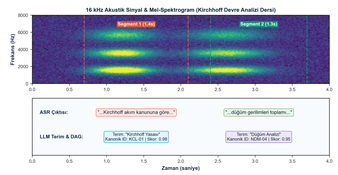


Figure 2: Acoustic Mel Spectrogram and Timestamped Term Extraction.

40-VIDEO EDUCATIONAL CORPUS

- **12 Disciplines:** STEM, medicine, social sciences, and corporate compliance (KVKK/GDPR, OHS). 7-lecture verified core test set.
- **Inter-Annotator Agreement:** Double-annotation on $N = 115$ items yields Cohen's $\kappa_R = 0.782$ (context) and $\kappa_D = 1.000$ (definition).

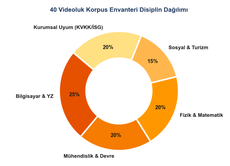


Figure 3: Corpus Discipline Distribution.

MULTILINGUAL ASR COMPARISON (WER)

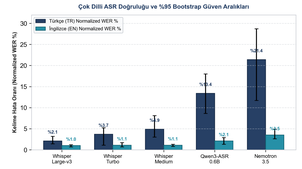


Figure 4: Normalized WER and 95% Confidence Intervals ($B = 10\,000$).

- **Whisper Large-v3:** Peak accuracy with 2.1% WER (TR) and 1.0% (EN).
- **Whisper Turbo:** High-speed RTF 0.025–0.044 (23–$40\times$ real-time).
- **VAD Degradation:** Energy-based VAD clips phonetic boundaries, degrading WER from 2.1% to 4.1%.

TABLE 1: INFERENCE TIME COMPARISON

Model / Hardware	TR Time	EN Time	Platform
EVREN Qwen 3.8F	**6.11 s**	**4.23 s**	National Cloud (H200)
Boğaziçi HPC (A40)	**15.28 s**	**12.45 s**	7B LoRA (48GB A40)
EVREN DeepSeek	27.68 s	24.12 s	High Throughput
Qwen 3.5-4B (Local)	39.80 s	34.20 s	RTX 5060 (4-bit NF4)
Qwen 3.5-9B (Local)	513.19 s	322.29 s	CPU Offload

BOĞAZİÇİ HPC A40 7B LoRA TRAINING

- **Parameter-Efficient Adaptation:** `Qwen2.5-7B-Instruct` adapted with $r = 16$, $\alpha = 32$ on A40 (Job ID `33580`).
- **Resource Profile:** 3 epochs in **13.67 s**; peak VRAM **27.84 GB**.
- **Cross-Domain Gain:** CS/BST recall rose from 0.0% to **33.3%**; Ateşman readability improved by **+25.7 points** into B1.

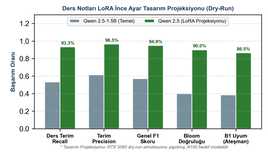


Figure 5: Qwen 2.5-7B LoRA Adaptation and Readability Gain.

HYBRID RAG AND PREREQUISITE GRAPH

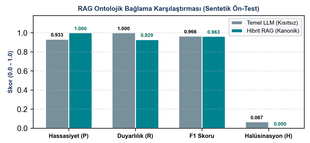


Figure 6: Hybrid RAG Grounding and Prerequisite Extraction.

- **Knowledge Grounding:** Contextualizes technical concepts across 7 domain ontologies via BM25 + Dense RRF fusion.
- **Prerequisite Gap Detection:** Detects missing prerequisite concepts skipped during rapid lecture delivery.

CROSS-LAYER ERROR PROPAGATION

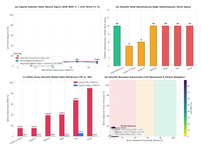


Figure 7: Empirical Error-Propagation Curves on HPC A40.

- **Acoustic Ladder:** 7 ASR levels tested (0% to 89.7% WER).
- **Sensitivity:** $\Delta\text{WER} = +5\% \implies \Delta F_1 = -0.11\%$ (TR) / -1.23% (EN).

FULL-CORPUS DAG AND PREREQUISITE GAPS

- **33 Lectures Analyzed:** Full corpus processed on A40 in 504.47 s ($\sim$8.4 min), extracting **118 technical terms** and topological dependency paths.
- **Pedagogical Gaps Identified:** 19 gaps detected (7 in KVKK, 5 each in OHS), highlighting concepts introduced without prerequisites.
- **CEFR-B1 Clarity:** **57.6%** of generated explanations fall in B1 band.



Figure 8: Concept Dependency Graph and Bloom Hierarchy.

TABLE 2: COMPETITIVE PLATFORM MATRIX

Feature	EduTerm AI	Coursera / edX	Duolingo
ASR Term Extraction	**Automated**	Manual Subtitles	Limited
Canonical RAG	**Yes (Hybrid)**	No	No
Prerequisite DAG	**Yes (NetworkX)**	Static Course Tree	Linear Tree
Edge / Private Inference	**Yes (Local/Web)**	Cloud Only	Cloud API
CEFR-B1 Scaffolding	**Yes (Ateşman)**	None	Fixed A1–B2

RESULTS AND CONCLUSION

- **Practical Viability:** Verified end-to-end multi-tier pipeline on Boğaziçi HPC (A40) and local RTX 5060; SLMs enable private edge tutoring.
- **Future Work:** Real-time LMS integration (Canvas/Moodle) and continuous student BKT knowledge tracing.

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